



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to PNG FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

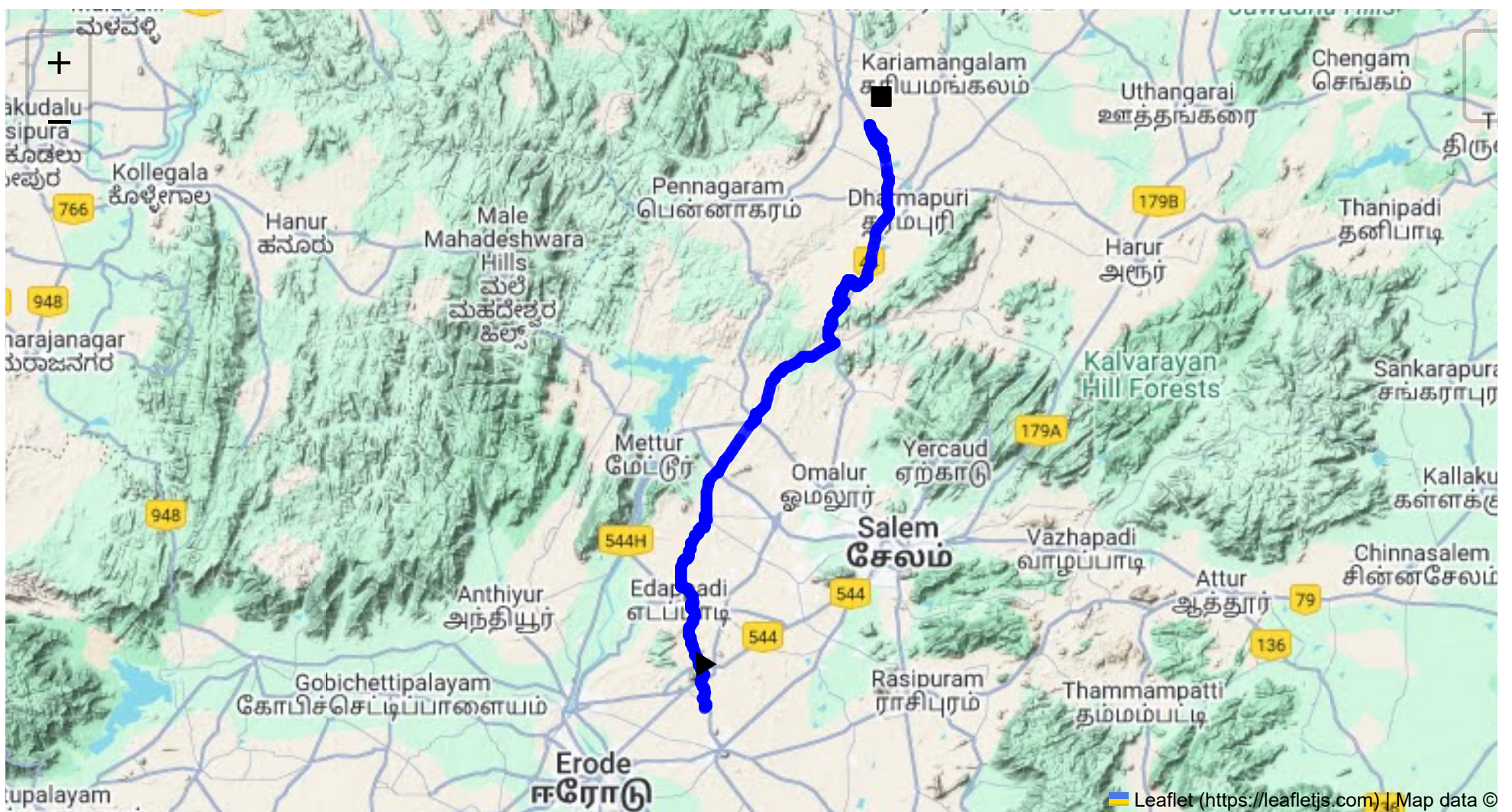
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 114.37 km
Estimated Duration: 2.6 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours
Start: (11.4381, 77.8734)
End: (12.212787, 78.119247)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Pulikkarai is approximately 114.37 kilometers, primarily covering rural and semi-urban areas of Tamil Nadu. It includes a combination of state highways and rural roads. Key waypoints involve passing through several small towns and villages, providing a diverse driving experience with varying road conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- **Weather Patterns:** This region typically experiences a tropical climate. Summers are hot, with temperatures peaking from March to June. Monsoon season (June to September) sees significant rainfall, which can lead to flooding and slippery roads.
- **Hazards:** During monsoons, roads may have water-logging issues, reduced visibility due to heavy rains, and landslides in certain areas. Summer heat can also affect vehicle performance and increase the risk of tire blowouts.

3. Analysis of Traffic Patterns

- **Peak Hours:** Typical business hours see increased traffic, especially in and around semi-urban areas like Sankari and Nallampalli.
- **Congestion-Prone Areas:** Expect traffic congestion near market zones and town centers, particularly Sankari and Dokkubodanahalli. Weekends may also have increased local traffic.

4. Assessment of Road Quality and Infrastructure

- **Road Quality:** Varies significantly; some parts have well-maintained highways, whereas others are narrow, with potholes and poor signage.
- **Infrastructure:** Limited in rural stretches; services and gas stations are more frequent near towns. Some bridges or culverts may be older and require caution.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, consider detouring through major highways:

- **Alternative Route:** Use the NH 544 and NH 44 where possible, as these are broader, better-maintained roads with quicker access to emergency services.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Regulations:** Restrictions may apply during certain hours in populous areas. Drivers must have valid permits for transporting hazardous materials and adhere to safety protocols.

7. Overview of Historical Incidents

Though specific data might not be readily available, local reports suggest occasional incidents involve overturned heavy vehicles, primarily caused by weather conditions or poor road surfaces.

8. Environmental Considerations and Sensitive Areas

- **Sensitive Areas:** Includes agricultural lands and forested areas. Care should be taken to avoid spillage of hazardous materials.

- **Environmental Impact:** Compliance with local waste disposal and emissions regulations is critical to protect local ecology.

9. Analysis of Communication Coverage

- **Coverage:** Generally good in towns but may be patchy in rural or heavily forested areas. Dead zones may occur around Kurumbapatti and Maniathahalli.
- **Suggestions:** Drivers should be equipped with alternative communication devices such as satellite phones for emergencies.

10. Estimated Emergency Response Times

- **In Towns:** Approximately 15-30 minutes due to proximity to local fire stations and medical facilities.
- **Rural Stretches:** Could extend to 45 minutes or more due to sparse facilities.

12. Overall Summary of Risk Assessment

- **Risks:** Moderate to high risks include weather-related hazards, variable road conditions, and congestion in populated areas.
- **Safety Measures:** Drivers should be adequately trained for emergency handling. Vehicles must be equipped with necessary safety gear and communication tools.
- **Mitigation:** Plan travel during daylight to avoid night-time challenges. Regular vehicle maintenance checks are essential, especially during extreme weather.

By understanding these factors, truck drivers can navigate the route more safely, minimizing risks associated with transporting hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60359, 77.83965	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	High	11.45028, 77.87202	15 KM/Hr
6	Turn	Medium	11.46400, 77.87323	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.47468, 77.86913	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	Medium	11.47934, 77.87008	30 KM/Hr
10	Turn	Medium	11.48045, 77.87147	30 KM/Hr
11	Turn	High	11.48277, 77.87250	15 KM/Hr
12	Turn	Medium	11.49490, 77.86661	30 KM/Hr
13	Turn	Medium	11.50909, 77.86025	30 KM/Hr
14	Turn	Medium	11.52561, 77.85522	30 KM/Hr
15	Turn	Medium	11.52766, 77.85585	30 KM/Hr
16	Turn	Medium	11.54721, 77.85263	30 KM/Hr
17	Turn	Medium	11.55446, 77.85619	30 KM/Hr
18	Turn	Medium	11.55636, 77.85791	30 KM/Hr
19	Turn	Medium	11.56395, 77.85969	30 KM/Hr
20	Turn	High	11.57903, 77.85432	15 KM/Hr
21	Turn	Medium	11.58067, 77.85762	30 KM/Hr
22	Turn	High	11.60434, 77.84010	15 KM/Hr
23	Turn	Medium	11.60978, 77.84165	30 KM/Hr
24	Turn	Medium	11.61795, 77.84020	30 KM/Hr
25	Turn	Medium	11.63085, 77.84072	30 KM/Hr
26	Turn	Medium	11.64073, 77.84627	30 KM/Hr
27	Turn	Medium	11.64130, 77.84816	30 KM/Hr
28	Turn	Medium	11.65274, 77.85197	30 KM/Hr
29	Turn	High	11.67310, 77.86031	15 KM/Hr
30	Turn	Medium	11.67357, 77.86193	30 KM/Hr
31	Turn	Medium	11.68196, 77.86690	30 KM/Hr
32	Turn	High	11.68492, 77.86898	15 KM/Hr
33	Turn	High	11.68463, 77.87262	15 KM/Hr
34	Turn	High	11.69455, 77.87583	15 KM/Hr
35	Turn	High	11.69558, 77.87439	15 KM/Hr
36	Turn	High	11.76019, 77.89047	15 KM/Hr
37	Turn	High	11.75964, 77.89200	15 KM/Hr
38	Turn	Medium	11.83309, 77.94416	30 KM/Hr
39	Turn	Medium	11.83367, 77.94485	30 KM/Hr
40	Turn	Medium	11.84157, 77.94713	30 KM/Hr
41	Turn	Medium	11.89721, 77.98056	30 KM/Hr
42	Turn	Medium	11.91887, 78.02599	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
43	Turn	Medium	11.93607, 78.05375	30 KM/Hr
44	Turn	High	11.93682, 78.05408	15 KM/Hr
45	Turn	Medium	11.95551, 78.04706	30 KM/Hr
46	Turn	High	11.95810, 78.04971	15 KM/Hr
47	Turn	Medium	11.96072, 78.04963	30 KM/Hr
48	Turn	Medium	11.96281, 78.05124	30 KM/Hr
49	Turn	High	11.96384, 78.05141	15 KM/Hr
50	Turn	High	11.96557, 78.04930	15 KM/Hr
51	Turn	High	11.96657, 78.04930	15 KM/Hr
52	Turn	High	11.99277, 78.06779	15 KM/Hr
53	Turn	Medium	11.99367, 78.06668	30 KM/Hr
54	Turn	Medium	11.99395, 78.06379	30 KM/Hr
55	Turn	Medium	11.99323, 78.06246	30 KM/Hr
56	Turn	High	11.99373, 78.06004	15 KM/Hr
57	Blind Spot	Blind Spot	12.00356, 78.06516	10 KM/Hr
58	Turn	High	12.00435, 78.06380	15 KM/Hr
59	Turn	High	12.00792, 78.06546	15 KM/Hr
60	Turn	High	12.00788, 78.06599	15 KM/Hr
61	Turn	Medium	12.01065, 78.06722	30 KM/Hr
62	Turn	High	12.01974, 78.07392	15 KM/Hr
63	Turn	High	12.02085, 78.07318	15 KM/Hr
64	Turn	High	12.02353, 78.07513	15 KM/Hr
65	Turn	High	12.02349, 78.07557	15 KM/Hr
66	Turn	High	12.02328, 78.07564	15 KM/Hr
67	Turn	Medium	12.02139, 78.08167	30 KM/Hr
68	Turn	Medium	12.02050, 78.08237	30 KM/Hr
69	Turn	Medium	12.01982, 78.08248	30 KM/Hr
70	Turn	Medium	12.01914, 78.08227	30 KM/Hr
71	Turn	Medium	12.01554, 78.08405	30 KM/Hr
72	Turn	High	12.01387, 78.08668	15 KM/Hr
73	Turn	Medium	12.12075, 78.13278	30 KM/Hr
74	Blind Spot	Blind Spot	12.23511, 78.10444	10 KM/Hr
75	Turn	High	12.23521, 78.10486	15 KM/Hr
76	Blind Spot	Blind Spot	12.21255, 78.11993	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
7	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
9	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
10	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
11	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
12	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
13	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
15	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
16	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
17	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
18	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
19	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
20	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
21	hospital	Selvakumar Hospital	11.836215, 77.945842	30 km/h	Medium
22	clinic	SK Clinic	11.83619, 77.9457752	30 km/h	Medium
23	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium
24	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60359, 77.83965



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46400, 77.87323



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47468, 77.86913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47934, 77.87008



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48045, 77.87147



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48277, 77.87250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49490, 77.86661



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50909, 77.86025



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.52561, 77.85522



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.52766, 77.85585



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54721, 77.85263



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55446, 77.85619



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55636, 77.85791



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56395, 77.85969



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57903, 77.85432



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58067, 77.85762



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60434, 77.84010



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60978, 77.84165



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61795, 77.84020



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63085, 77.84072



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64073, 77.84627



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64130, 77.84816



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65274, 77.85197



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67310, 77.86031



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67357, 77.86193



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68196, 77.86690



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68492, 77.86898



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69455, 77.87583



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69558, 77.87439



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76019, 77.89047



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75964, 77.89200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.83309, 77.94416



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83367, 77.94485



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.84157, 77.94713



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.89721, 77.98056



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91887, 78.02599



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93607, 78.05375



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.93682, 78.05408



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95551, 78.04706



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95810, 78.04971



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96072, 78.04963



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96281, 78.05124



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96384, 78.05141



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96557, 78.04930



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96657, 78.04930



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.99277, 78.06779



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.99367, 78.06668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99395, 78.06379

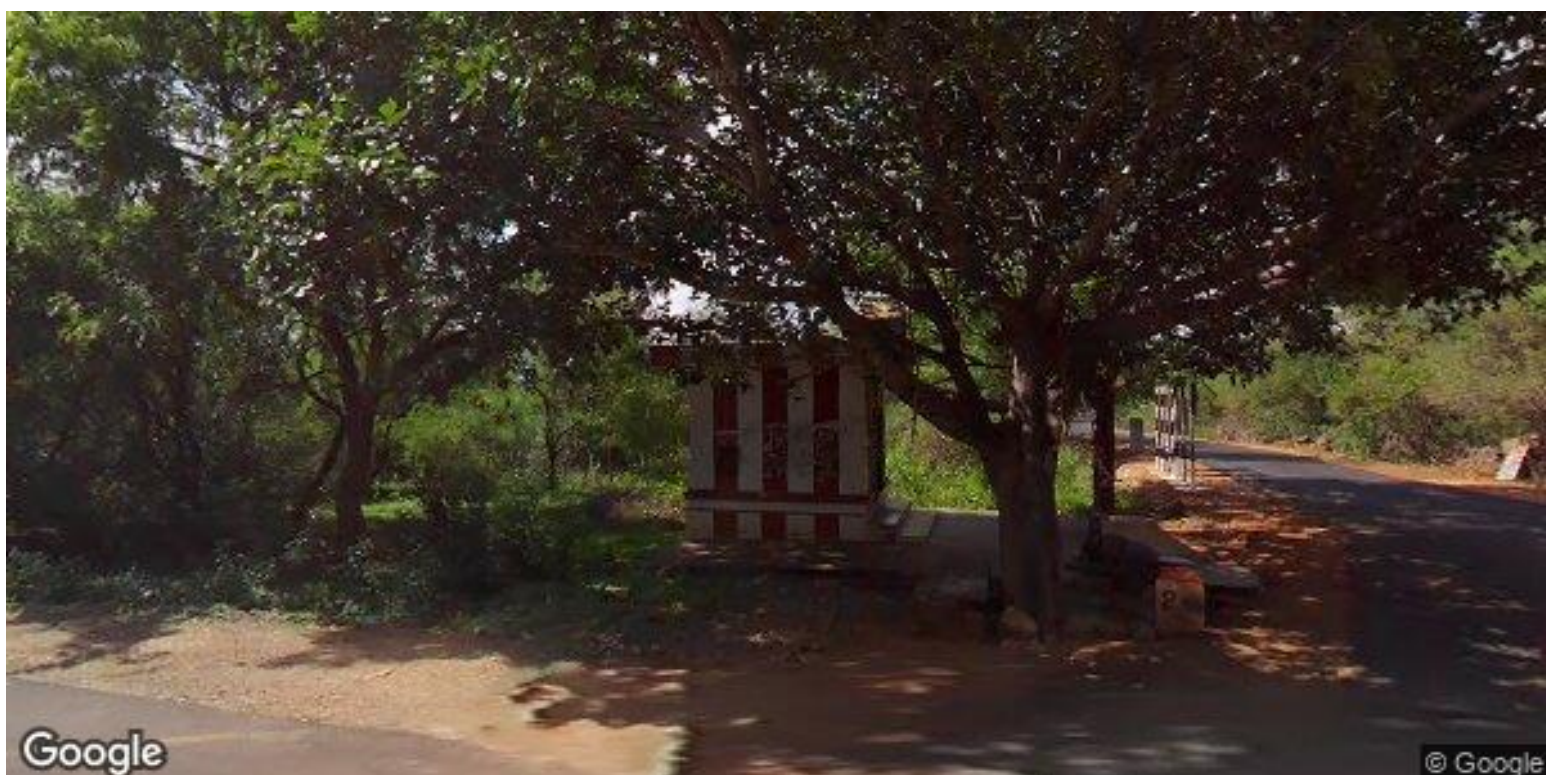


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99323, 78.06246



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.99373, 78.06004



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.00356, 78.06516



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.00435, 78.06380



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.00792, 78.06546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.00788, 78.06599



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.01065, 78.06722



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.01974, 78.07392



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.02085, 78.07318



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.02353, 78.07513



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.02349, 78.07557



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.02328, 78.07564



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.02139, 78.08167



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.02050, 78.08237



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.01982, 78.08248



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.01914, 78.08227



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.01554, 78.08405



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.01387, 78.08668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.12075, 78.13278

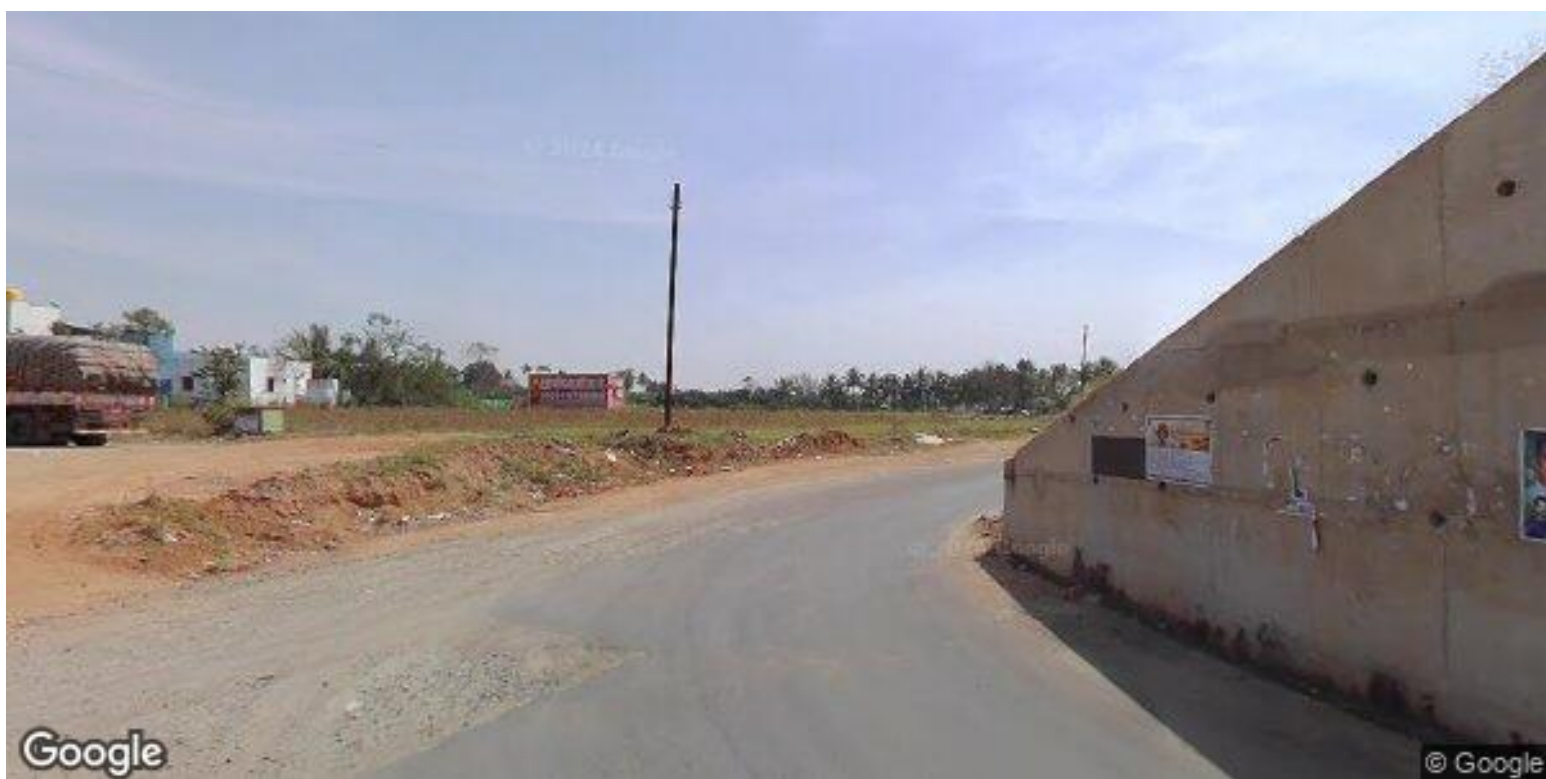


Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.23511, 78.10444



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.23521, 78.10486



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.21255, 78.11993
